

ASSGINMENT : 5 – TYPES OF OPERATING SYSTEM

GUIDED BY : VIPUL MISTRI

Q.1 Create a table comparing at least three types of operating systems (such as Windows, Linux, and Android) based on their main features, target devices, and examples of apps you use on each.

ANS :

Operating System	Main Features	Target Devices	Example Apps
Windows	Wide software compatibility, user-friendly graphical interface, strong legacy hardware support, comprehensive gaming features (DirectX).	Desktop PCs, Laptops, Workstations, 2-in-1 Tablets.	Microsoft Office, Adobe Photoshop, Cisco Packet Tracer, Steam games.
Linux	Open-source & customizable, light resource usage, robust terminal/CLI, built-in security features, multi-user architecture.	Servers, Desktop PCs, Laptops, Cloud infrastructure, Embedded systems.	Wireshark, GIMP, VS Code, LibreOffice, Terminal tools.
Android	Open-source mobile OS (Linux-based), optimized for touchscreens, deep integration with Google services, extensive app ecosystem via Google Play.	Smartphones, Tablets, Smart TVs, Smartwatches, Automotive units.	Zomato, Swiggy, PhonePe, WhatsApp, Instagram.

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Q.2 Identify the operating system running on your phone and your laptop/desktop. Write down two unique features of each OS that affect your daily use of popular apps like Instagram or Zomato.

ANS : Here are the operating systems running on my primary devices, along with two unique OS features that directly impact how popular apps like Instagram or Zomato are used:

1. Mobile OS: Android

- **App Pairing & Split-Screen View:** Android allows you to pair two apps together to launch simultaneously in split-screen mode.
 - *Impact on Apps:* You can run **Zomato** on the top half of your screen to track a live food delivery order while simultaneously scrolling through **Instagram** on the bottom half, without needing to constantly switch back and forth.
- **Notification Channels & Priority Alerts:** Android allows granular control over app notifications, letting you toggle specific types of alerts (e.g., promotional offers vs. live delivery updates).
 - *Impact on Apps:* You can mute noisy promotional discounts from **Zomato** while ensuring real-time delivery status alerts and driver calls always pop up on your lock screen.

2. Laptop/Desktop OS: Windows

- **Snap Layouts (Window Management):** Windows 11 lets you instantly snap multiple open application windows into organized grid layouts.
 - *Impact on Apps:* If you open the **Zomato** web portal in a browser alongside a work document, you can lock Zomato to a small side panel to monitor order progress while dedicating the rest of your screen space to active work.
- **Phone Link Integration:** Microsoft Phone Link syncs your Android or iOS smartphone notifications, calls, and app alerts directly to your desktop environment.
 - *Impact on Apps:* When you place a food order on **Zomato** or receive an **Instagram** direct message on your phone, the notification instantly appears on your Windows desktop, allowing you to view alerts or reply to messages without picking up your phone.

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Q.3 List three security risks that are specific to mobile operating systems like Android or iOS, and explain how these risks could impact apps you use (for example, WhatsApp or Paytm).

ANS : Three security risks specific to mobile operating systems (Android/iOS) include:

1. Malicious Screen Overlays (Overlay Attacks / Tapjacking)

- **The Risk:** Mobile OS architectures allow apps to draw over other apps (e.g., using Android's `SYSTEM_ALERT_WINDOW` permission or accessibility APIs). Malicious background apps use this feature to display transparent or duplicate UI windows directly above legitimate financial or messaging apps.
- **Impact on Apps:** When you open an app like **Paytm**, the hidden malware can place an invisible overlay over the password or PIN field to steal your credentials as you type. It can also trick you into clicking an invisible "Confirm Transfer" button under the guise of an innocent pop-up button.

2. Excessive & Over-Privileged OS Permissions

- **The Risk:** Mobile OS environments rely heavily on granular permission models (SMS, Contacts, Camera, Accessibility Services, Location). Attackers create simple utility apps (like a flashlight or calculator) that request dangerous runtime permissions.
- **Impact on Apps:** If a malicious app gets access to read your SMS messages, it can intercept incoming One-Time Passwords (OTPs) sent by **Paytm** or **WhatsApp**. This allows attackers to bypass two-factor authentication, gain control of your WhatsApp account on another device, or approve unauthorized money transfers.

3. Untrusted App Sideloaded & Malicious APKs/IPAs

- **The Risk:** Unlike traditional desktops with open installation models, mobile operating systems restrict app installations to vetted app stores by default. Enabling "Install from Unknown Sources" (sideloading) bypasses OS sandbox checks and malware scanning mechanisms provided by Google Play Store or Apple App Store.
- **Impact on Apps:** Scammers frequently distribute modified or "cracked" versions of apps (e.g., a fake WhatsApp MOD promising extra customization features) via third-party websites or messaging groups. Installing a sideloaded, modified APK can result in modified app code that

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quietly steals private chat logs from **WhatsApp** or reads banking credentials stored in local storage.

Q.4 Research and describe how a cloud-based operating system (such as Chrome OS or a virtual desktop environment) is different from a traditional desktop OS. Give one example of how this difference changes the way you access apps like YouTube or Flipkart.

ANS : A **cloud-based operating system** (like Chrome OS or a Virtual Desktop Infrastructure/VDI) relies on web technologies and remote servers to run applications and process data, whereas a **traditional desktop OS** (like Windows or macOS) runs applications, manages files, and processes computational workloads locally on physical hardware.

Here are the key differences and an example of how this alters daily app access:

Core Differences

1. Storage & Execution:

- **Traditional OS:** Installs software applications directly onto local storage drives (HDD/SSD) and uses local hardware (CPU/RAM) to execute them.
- **Cloud OS:** Uses web-based applications (Progressive Web Apps/PWAs) or streams virtualized applications from remote servers over the internet.

2. Hardware Dependence & Recovery:

- **Traditional OS:** Heavily reliant on local hardware components. If your hard drive fails or your computer crashes, local files and offline software may be lost unless manually backed up.

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- **Cloud OS:** Lightweight and hardware-independent. All user profile settings, application preferences, and files are saved continuously to the cloud, allowing seamless login to a fresh device with zero data loss.

3. Maintenance & Updates:

- **Traditional OS:** Requires periodic software patches, manual driver updates, and antivirus updates installed directly onto the machine.
- **Cloud OS:** Updates automatically in the background in seconds with minimal system downtime, relying on server-side sandboxing and continuous cloud updates for security.

How This Changes Access to Apps (Example: YouTube or Flipkart)

- **On a Traditional OS:** You access **YouTube** or **Flipkart** primarily by opening a desktop web browser or running a heavy standalone desktop app installed on your drive. If your local network interface card or storage fails, the local app fails to function.
- **On a Cloud OS:** You access **Flipkart** or **YouTube** as a light **Progressive Web App (PWA)** or direct web portal pinned to your shelf. Because the processing, video streaming decoding optimization, and user state are managed cloud-side, clicking the icon opens an instant, app-like experience without downloading large installation packages. If your device dies mid-stream, opening a new device lets you pick up your video or shopping cart instantly where you left off.